

LA GRANDE RIVIERE, QC, Canada

WMO: 718270

Lat: 53.630N

Lon: 77.700W

Elev: 195

StdP: 99.01

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -34.5	(c) -32.5	(d) -39.0	(e) 0.1	(f) -34.4	(g) -37.0	(h) 0.1	(i) -32.4	(j) 11.2	(k) -17.5	(l) 10.0	(m) -17.6	(n) 3.1	(o) 200	(p) 0.801

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	26.8	17.3	24.7	16.1	22.7	15.3	19.0	24.0	17.8	22.0	16.6	20.6	5.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.4	12.7	20.5	16.1	11.8	19.4	15.0	10.9	18.5	55.1	23.8	51.1	21.9	47.5	20.6	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.4	9.2	8.3	-36.9	30.5	2.3	2.2	-38.5	32.0	-39.9	33.3	-41.1	34.6	-42.8	36.2
			-37.0	21.0	2.3	1.4	-38.6	22.1	-40.0	22.9	-41.2	23.7	-42.9	24.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.8	-21.4	-20.7	-13.5	-4.6	4.7	11.7	14.8	13.7	8.9	2.9	-4.9	-14.1
	DBStd	13.92	7.25	6.97	7.92	6.51	6.13	5.45	4.09	3.92	4.20	4.09	4.78	7.03
	HDD10.0	4738	973	858	728	439	191	44	7	8	70	224	448	747
	HDD18.3	7373	1232	1092	987	689	425	207	124	152	283	479	698	1006
	CDD10.0	441	0	0	0	0	25	96	154	124	38	4	0	0
	CDD18.3	34	0	0	0	0	1	8	14	9	1	0	0	0
	CDH23.3	336	0	0	0	0	25	103	131	70	7	0	0	0
	CDH26.7	66	0	0	0	0	4	22	29	11	0	0	0	0
(14) Wind	WSAvg	4.2	4.0	3.9	4.1	4.2	4.2	4.2	4.0	4.1	4.5	4.4	4.4	4.1
(15) Precipitation	PrecAvg	717	33	21	28	29	45	62	86	100	107	90	72	44
	PrecMax	893	87	44	48	59	92	128	146	162	234	158	140	73
	PrecMin	557	7	9	4	6	14	28	32	41	50	54	29	23
	PrecStd	90	19	10	12	14	21	29	29	32	36	24	20	14
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.2	0.4	7.1	14.2	26.9	29.5	29.5	28.3	24.3	17.3	8.4	1.5
		MCWB	-0.2	-0.4	3.5	6.9	15.2	19.2	18.7	18.6	17.4	12.7	6.9	0.9
	2%	DB	-3.2	-3.0	3.6	9.4	22.1	26.2	26.6	25.3	20.6	13.7	4.7	-0.2
		MCWB	-3.6	-3.9	1.0	4.4	13.0	16.3	17.3	17.4	15.4	11.2	3.8	-0.5
	5%	DB	-6.8	-5.6	1.1	6.8	18.5	23.8	24.3	22.9	18.1	11.3	2.7	-2.3
		MCWB	-7.2	-6.3	-0.5	2.7	11.3	15.2	16.0	16.4	14.3	9.4	2.0	-2.7
	10%	DB	-10.2	-9.4	-1.2	4.8	15.2	21.3	22.3	20.6	15.8	8.9	1.0	-4.3
		MCWB	-10.6	-10.1	-2.7	1.5	9.3	13.8	15.4	15.3	12.8	7.3	0.5	-4.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.2	-0.2	4.3	7.3	16.3	20.2	21.1	20.1	18.6	14.1	7.3	1.0
		MCDB	0.0	0.4	6.8	12.8	23.2	27.5	26.8	24.2	22.4	16.5	8.1	1.3
	2%	WB	-3.7	-3.8	1.4	4.9	13.8	18.4	18.7	18.7	16.7	11.8	3.9	-0.6
		MCDB	-3.2	-3.1	3.3	8.7	20.4	23.4	23.4	23.3	19.3	13.4	4.5	-0.3
	5%	WB	-7.1	-6.3	-0.4	3.2	11.7	16.4	17.5	17.5	15.0	9.6	2.0	-2.6
		MCDB	-6.8	-5.6	1.1	6.4	18.0	21.2	21.6	21.5	17.4	11.1	2.6	-2.3
	10%	WB	-10.6	-10.1	-2.8	1.7	9.6	14.7	16.4	16.2	13.0	7.5	0.5	-4.7
		MCDB	-10.2	-9.4	-1.2	4.3	14.6	20.5	20.7	19.6	15.5	8.9	1.0	-4.3
(35) Mean Daily Temperature Range	MDBR		8.1	9.8	11.1	10.1	10.9	12.0	10.7	9.5	7.9	5.7	5.2	6.6
	5% DB	MCDBR	9.6	10.7	10.7	11.3	15.3	14.4	13.3	12.3	11.2	9.6	5.8	6.8
		MCWBR	9.6	10.2	9.1	7.5	8.5	7.3	6.5	6.0	7.3	5.4	6.9	
	5% WB	MCDBR	9.6	10.7	10.9	10.6	14.6	12.7	11.2	10.4	10.2	8.8	5.7	6.7
		MCWBR	9.7	10.3	9.6	7.4	8.8	8.1	6.7	6.2	7.7	7.4	5.5	6.8
(40) Clear-Sky Solar Irradiance	taub		0.245	0.264	0.280	0.329	0.346	0.356	0.370	0.359	0.347	0.305	0.273	0.225
	taud		2.169	2.192	2.210	2.195	2.396	2.428	2.424	2.457	2.472	2.540	2.289	2.208
	Ebn at Noon		729	839	905	898	902	893	872	863	829	801	690	690
	Edh at Noon		63	87	108	126	110	108	107	97	85	64	57	49
(44) All-Sky Solar Radiation	RadAvg		0.93	2.04	3.63	5.06	5.49	5.88	5.17	4.06	2.73	1.51	0.68	0.57
	RadStd		0.10	0.16	0.32	0.37	0.42	0.52	0.41	0.32	0.28	0.13	0.09	0.08

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-13
(47) Regional (1 neighbor)	N/A	N/A	+1.16	-0.72	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page